

Electronic Commerce: Virtual Corporations Selling to a Virtual Marketplace

Aberdeen Group research indicates the Information Technology (IT) industry is about to wrestle with the radically changing marketplace brought about by the Internet. Basic business processes such as buying and selling that have evolved over centuries are going to change dramatically as both buyers and sellers demand more and better information, faster, and for competitive advantage. This sea change — a true paradigm shift — will forever alter the way society operates and how IT is utilized. For the IT executives, aligning IT infrastructure with fast evolving electronic-commerce business practices will be difficult at best. This *White Paper* summarizes recent Aberdeen field research and our current best practices for electronic commerce initiatives.

Not One, But Three Kinds of Electronic Commerce

Many organizations still think of electronic commerce as one, monolithic business process. We suggest it's easier to discuss and plan for electronic commerce by looking at the three combinations of buyer and seller: business-to-business; business-to-consumer; and, consumer to consumer.

Business-to-Business: Shortening the Business Supply Chain

Business-to-Business electronic commerce looks at the wholesale side of business goods and services. To build your product or service, your company buys raw materials and contracted services. Orders tend to be regular and certainly regulated. Your customer may be another business or organization further along in what is called the "supply chain". The object of business-to-business electronic commerce is to make the supply chain as efficient as possible, by creating a virtual corporation along the supply chain. This must be sustained by IT that enables supply chain members to electronically exchange business documents (i.e., EDI) such as purchase orders and invoices, while anticipating demand through decision support systems (i.e., data warehousing). Moreover, business-to-business collaboration via e-mail and conferencing becomes a critical enabler. In fact, the Internet provides an ideal technology base for dealing with vertical-market business issues. Finally, the virtual corporation must efficiently harness internal resources too. This means a strong internal Intranet connecting employees and workgroups as well as Internet connections to suppliers.

Business-to-Consumer: They Want It Their Way

The good news about the virtual marketplace is that it truly does open up new selling opportunities around the globe. But "Just hanging out your company's sign on the Internet won't bring the customers streaming into your store" is the way one Aberdeen client characterized her experiences in business-to-consumer electronic commerce. Today's consumers become jaded in an Internet year — a few weeks — by generalized Internet commerce sites with static pages, complex menu hierarchies, and difficult purchase mechanisms. The key to business success going forward will be delivering fine-granularity individual personality to consumers — self-service. This means keeping detailed records about purchasing patterns such as color, garment sizes, shipping methods, coupons and promotions. A dynamically built catalog tuned to the consumer's

individual preferences is ideal. And please note that the MTV generation wants an exciting Internet commerce experience — audio, motion, cool graphics — while your father's generation is more comfortable with a more reassuring look and feel. Consumers too want to collaborate with their business suppliers, so Internet e-mail feedback and customer help-desk chat sessions can make the consumer feel more attached to their suppliers.

Consumer-to-Consumer: On-Line Classified Ads

The Internet will be the ultimate classified ad page. It will allow anyone to sell anything anywhere. With the exception of newspapers who face a dramatic market shift, the consumer-to-consumer EC market can be safely ignored by IT organizations.

Building or Buying an Electronic Commerce Architecture

Aberdeen's research with pioneering electronic organizations confirms the need for a technology-architected approach to Internet electronic commerce. In short, an architected approach is more likely to stand the test of Internet time, to deal with fast evolving yet presently imprecise technology such as distributed objects, and to glue disparate production systems together. The principles governing architectural choices should favor flexibility and expandability — the ability to change the system frequently as a result of market-driven demand and to cope with the business success brought on by high volumes. Moreover, Aberdeen's recommendations on implementing Internet electronic commerce have recently changed.

Until very recently, the technology available to efficiently build and maintain flexible, expandable — and compelling — EC systems was fragmented and not in common use. This virtually dictated that in-house IT organizations build their own EC infrastructure, with the obvious costs and risks of pioneering. Going forward, we recommend that IT organizations carefully evaluate the option of buying an integrated, architected solution for business-to-business or business-to-consumer electronic commerce. By buying an architected technology framework, IT organizations minimize implementation risks, can get systems to market faster, and will benefit from the suppliers' experience, support and training.

Internet Electronic Commerce Technologies

To many IT practitioners, Internet EC will remind them of the childhood homily about "something old, something new". In some cases such as databases, EC adds new technology onto an old base. And new technology such as payment authentication and encryption must be applied using the familiar rules of transaction processing in production-quality business systems. The key Internet EC technologies to consider are:

Security: authentication, encryption, firewalls, password control

Payment Systems: credit card processing, 3rd party credit authorization, electronic cash, EDI

Distributed Transaction Processing: session management, deferred commit, multi-server transactions

Universal Databases: add searchable new media to RDBMS

Store-Front Building: specialized application development tools

Collaboration: e-mail, work flow, document management, EDI
Development Tools: 4GL and CASE tools generating Java
Middleware: gateways, object request brokers, CGI replacements

Conclusions

Our electronic commerce research indicates significant IT investments totaling over \$7.2 billion are planned over the next three years for information technology used for Internet electronic commerce systems. During that time, the majority of business-to-consumer companies will roll out their first or second generation EC systems. Less visible — and hidden by the secrecy cloak of competitive advantage — will be business-to-business EC collaboration aimed at making the supply chain more efficient.

Technology *change* will be the hallmark of electronic commerce information technology over at least the next decade. Moreover, Internet EC transaction volumes will grow rapidly but not overnight — yet the exceptions will see explosive growth. These two well considered predictions by Aberdeen Group lead us to strongly recommend that IT practitioners carefully architect their EC endeavors, and consider whether to make or buy their core EC technology.

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